

# Winning Space Race with Data Science

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# Outline

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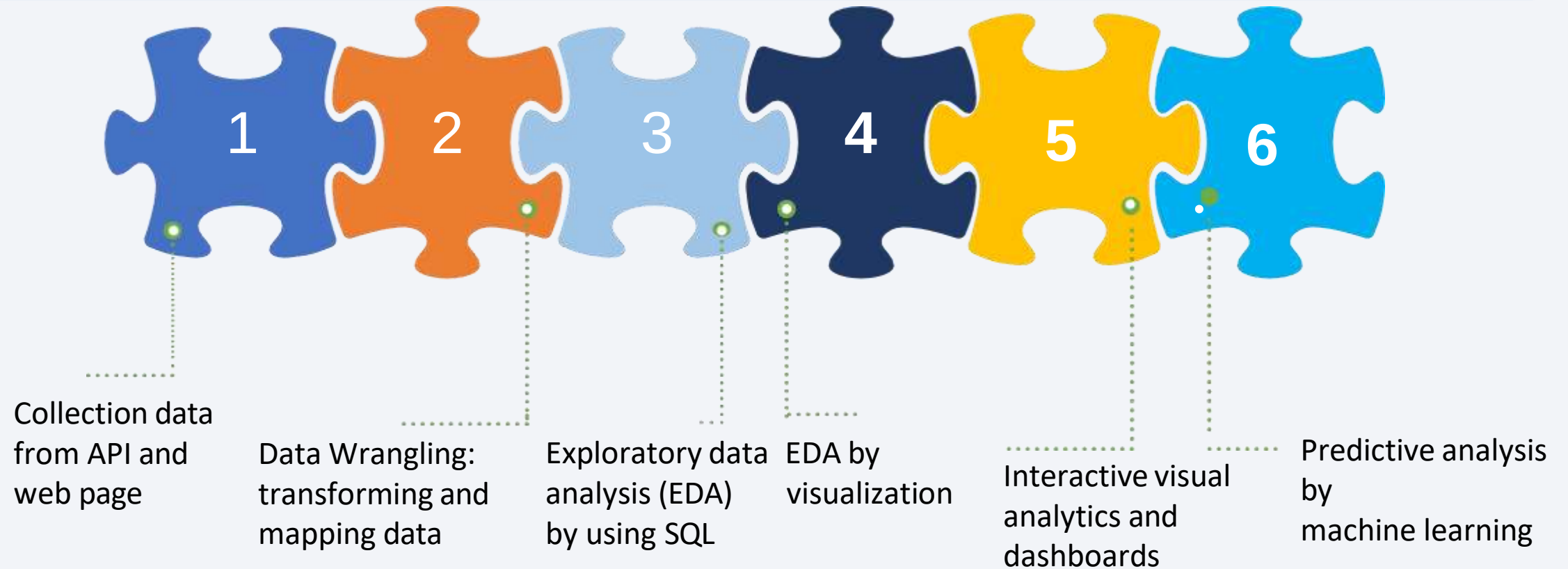
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# Executive Summary



*Summary: Select the best model for future prediction.*

# Introduction



## SpaceX's goal:

- Sending spacecraft to the International Space Station.
- Starlink, a satellite internet constellation providing satellite Internet access.
- Sending manned missions to Space.

### Object

SpaceX's Falcon 9 launch

### Goal

To determine the price of each launch.

### Reason

If we can determine if the first stage will land, we can determine the cost of a launch

### Method

By gathering information about Space X and creating dashboards:

- To determine if SpaceX will reuse the first stage
- Train a machine learning model and use public information to predict if SpaceX will reuse the first stage

Section 1

# Methodology

- Data collection methodology:
  - Require the data from SpaceX API
  - Collect data from a Wikipedia page
- Perform data wrangling
  - Perform EDA to find some patterns
  - Determine what would be the label for training supervised model

Perform exploratory data analysis (EDA) using visualization and SQL

- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Compare logistic regression model, support vector machine tree decision classifier, KNN by using GridSearchCV to select the best fit model

# Data Collection



## 1 AP

spacex\_url="<https://api.spacexdata.com/v4/launches/past>"

- Required the data from Space API
- Clean the data

## 2 Web

[https://en.wikipedia.org/wiki/List\\_of\\_Falcon\\_9\\_and\\_Falcon\\_Heavy\\_launches](https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches)

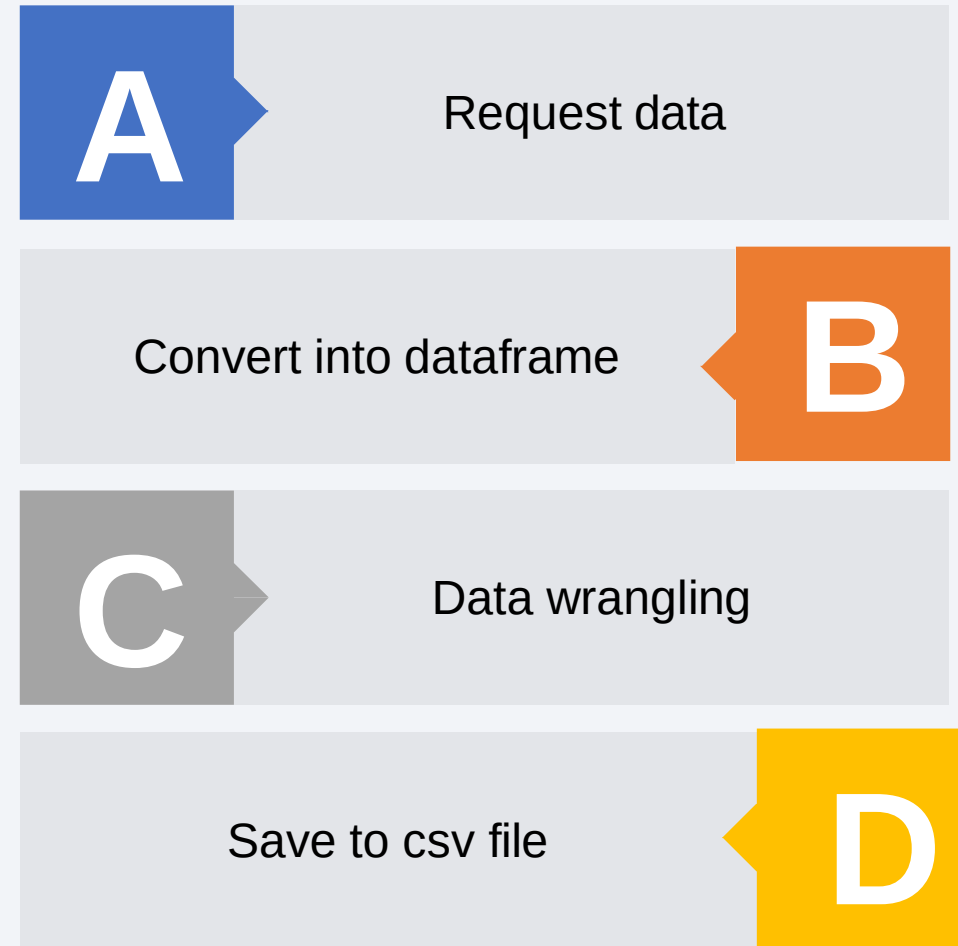
- Extract a Falcon 9 launch records HTML table from Wikipedia
- Parse the table and convert it into a Pandas data frame

# Data Collection - SpaceX API



- Request data from SpaceX API
- Convert the json result into a dataframe
- Filter dataframe to only Falcon 9 launches and data wrangling
- Export to csv
- Link :

<https://github.com/Jananijana2712/SpaceX-Falcon-9-First-Stage-Landing-Prediction/blob/main/1.1%20SpaceX%20the%20Data%20Collection%20API%20Lab.ipynb>

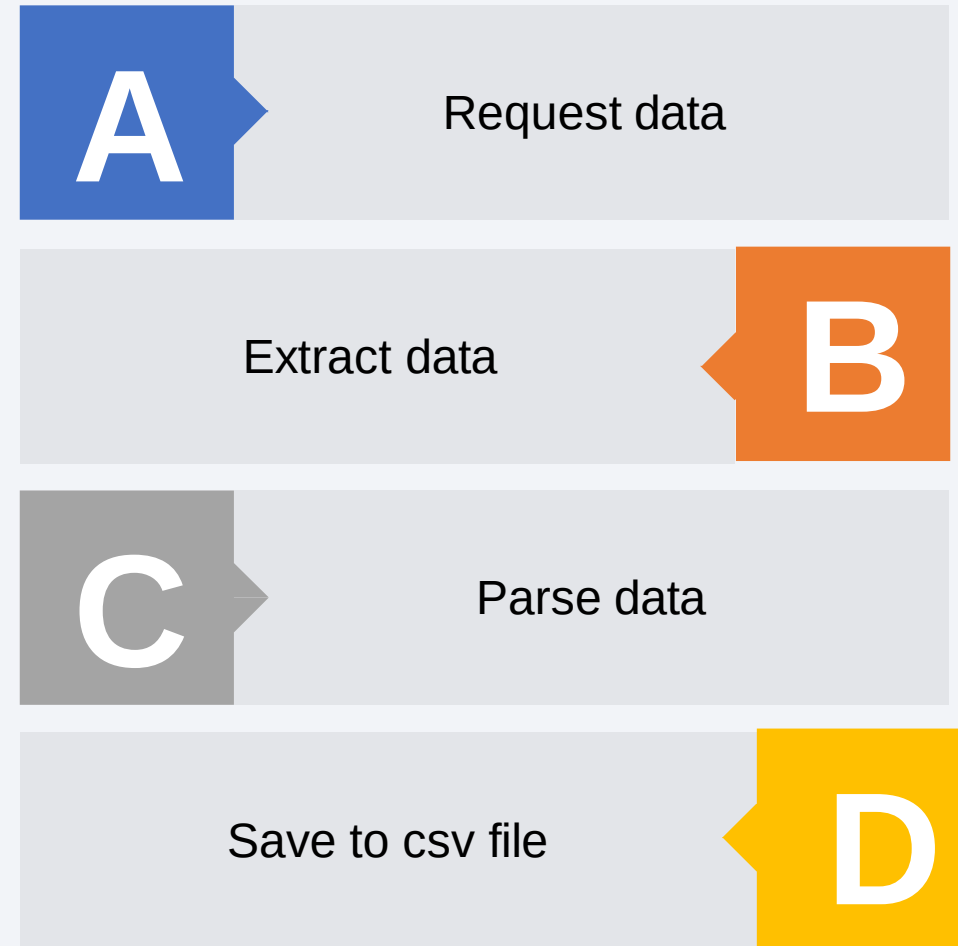


# Data Collection - Scraping



- Request the Falcon9 Launch Wiki page from its URL
- Extract all column/variable names from the HTML table header
- Create a data frame by parsing the launch HTML tables
- Export to csv
- Link :

[https://github.com/Jananijana2712/SpaceX-Falcon\\_9\\_First\\_Stage\\_Landing\\_Prediction/blob/main/1.2%20SpaceX%20the%20Data%20Collection%20with%20Web%20Scraping%20lab.ipynb](https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.2%20SpaceX%20the%20Data%20Collection%20with%20Web%20Scraping%20lab.ipynb)

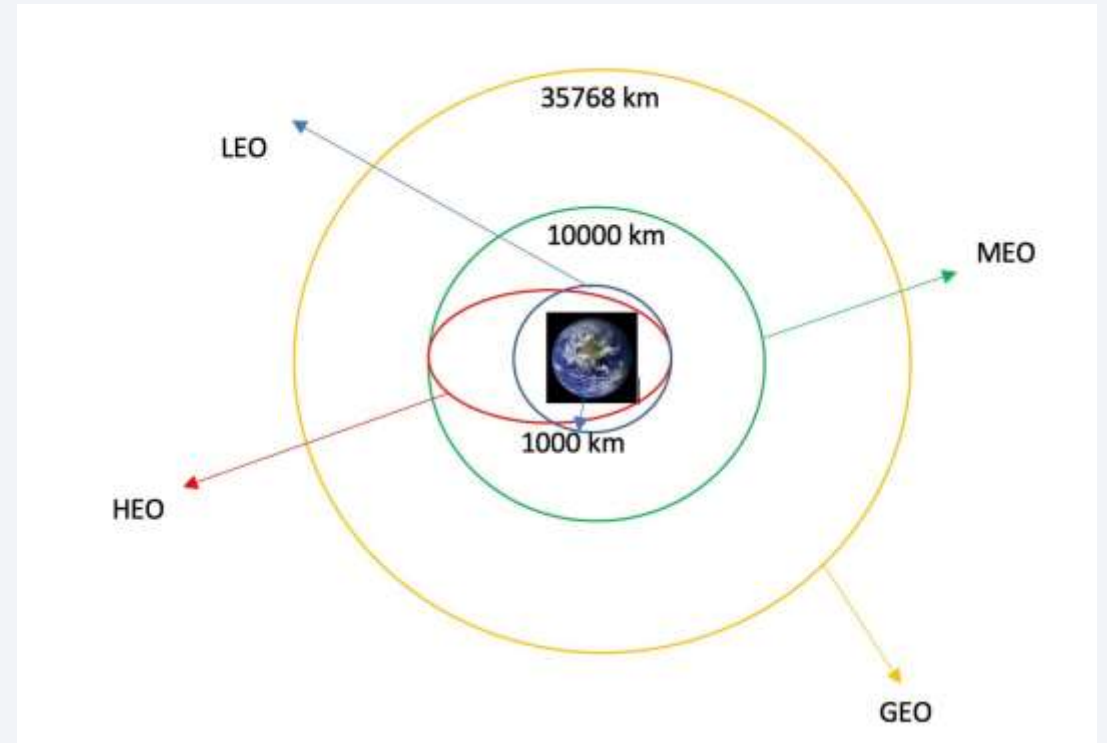


# Data Wrangling



- Specify the missing value
- Calculate the number of launches on each site
- Calculate the number and occurrence of each orbit
- Calculate the number and occurrence of each orbit
- Create a landing outcome label from Outcome column

Link: [https://github.com/Jananijana2712/SpaceX-Falcon\\_9\\_First\\_Stage\\_Landing\\_Prediction/blob/main/1.3%20SpaceX%20Data%20Wrangling.ipynb](https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.3%20SpaceX%20Data%20Wrangling.ipynb)



- The relationship between Flight Number and Launch Site -> scatter plot
  - The relationship between Payload and Launch Site -> scatter plot
  - The relationship between success rate of each orbit type -> bar plot
  - The relationship between Flight Number and Orbit type -> scatter plot
  - The relationship between Payload and Orbit type -> scatter plot
  - The launch success yearly trend -> line chart
- 
- The scatter plot is the best to describe the relation between two categorical data
  - The bar plot is the best to compare several categorical data
  - The line plot is the best to show the time series data



[https://github.com/Jananijana2712/SpaceX-Falcon\\_9\\_First\\_Stage\\_Landing\\_Prediction/blob/main/1.4%20SpaceX%20%20the%20EDA%20with%20Visualization.ipynb](https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.4%20SpaceX%20%20the%20EDA%20with%20Visualization.ipynb)

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first successful landing outcome in ground pad was achieved.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List the names of the booster\_versions which have carried the maximum payload mass.
- List the records which will display the month names, failure landing\_outcomes in drone ship ,booster versions, launch\_site for the months in year 2015.
- Rank the count of successful landing\_outcomes between the date 04-06-2010 and 20-03-2017 in descending order.



[https://github.com/Jananijana2712/SpaceX-Falcon\\_9\\_First\\_Stage\\_Landing\\_Prediction/blob/main/Assignment%20SQL%20Notebook%20for%20Peer%20Assignment.ipynb](https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/Assignment%20SQL%20Notebook%20for%20Peer%20Assignment.ipynb)

# Build an Interactive Map with Folium



- Mark all launch sites on a map
- Mark the success/failed launches for each site on the map
- Calculate the distances between a launch site to its proximities
  - Whether it is close to the coast
  - Whether it is close to the railway
  - Whether it is close to the highway
  - Whether it is close to the city

► **To find some geographical patterns about launch sites**



[https://github.com/Jananijana2712/SpaceX-Falcon\\_9\\_First\\_Stage\\_Landing\\_Prediction/blob/main/1.5%20SpaceX%20Interactive%20Visual%20Analytics%20with%20Folium.ipynb](https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.5%20SpaceX%20Interactive%20Visual%20Analytics%20with%20Folium.ipynb)

# Build a Dashboard with Plotly Dash



- A launch site drop-down input component
- A success-pie-chart based on the selected site dropdown
- A range slicer to select payload
- A success-payload-scatter-chart scatter plot based on the selected site dropdown

► **To inspect the relationship of success rate between launch site and payload**



[https://github.com/Jananijana2712/SpaceX-Falcon\\_9\\_First\\_Stage\\_Landing\\_Prediction/blob/main/spacex%20dash%20app.py](https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/spacex%20dash%20app.py)

# Predictive Analysis (Classification)



Data wrangling

Data standarization

Split into traning  
and test datasets

Predictive  
model  
evalutation

Predictive  
model  
selection



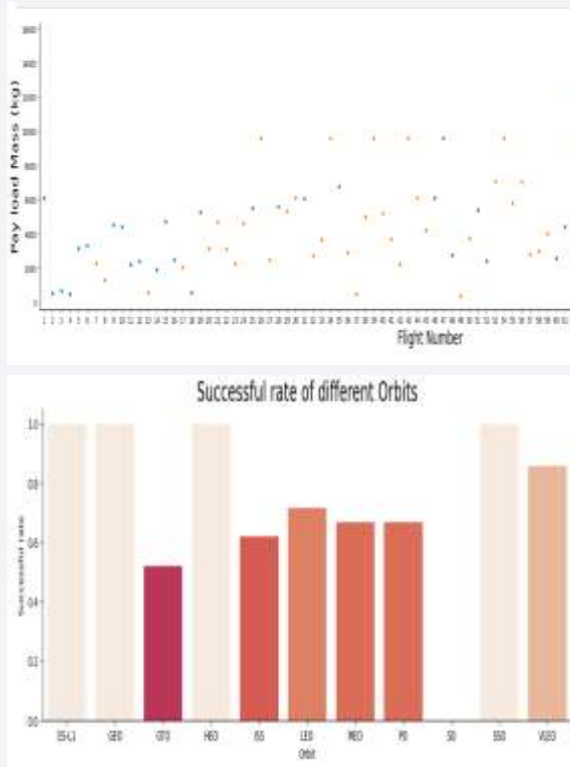
[https://github.com/Jananijana2712/SpaceX-Falcon\\_9\\_First\\_Stage\\_Landing\\_Prediction/blob/main/1.6%20SpaceX%20Machine%20Learning%20Prediction.ipynb](https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.6%20SpaceX%20Machine%20Learning%20Prediction.ipynb)

- Logistic regression
  - Support vector machine
  - Decision tree classifer
  - K-nearest neighbors
- K-nearest neighbors

## EDA

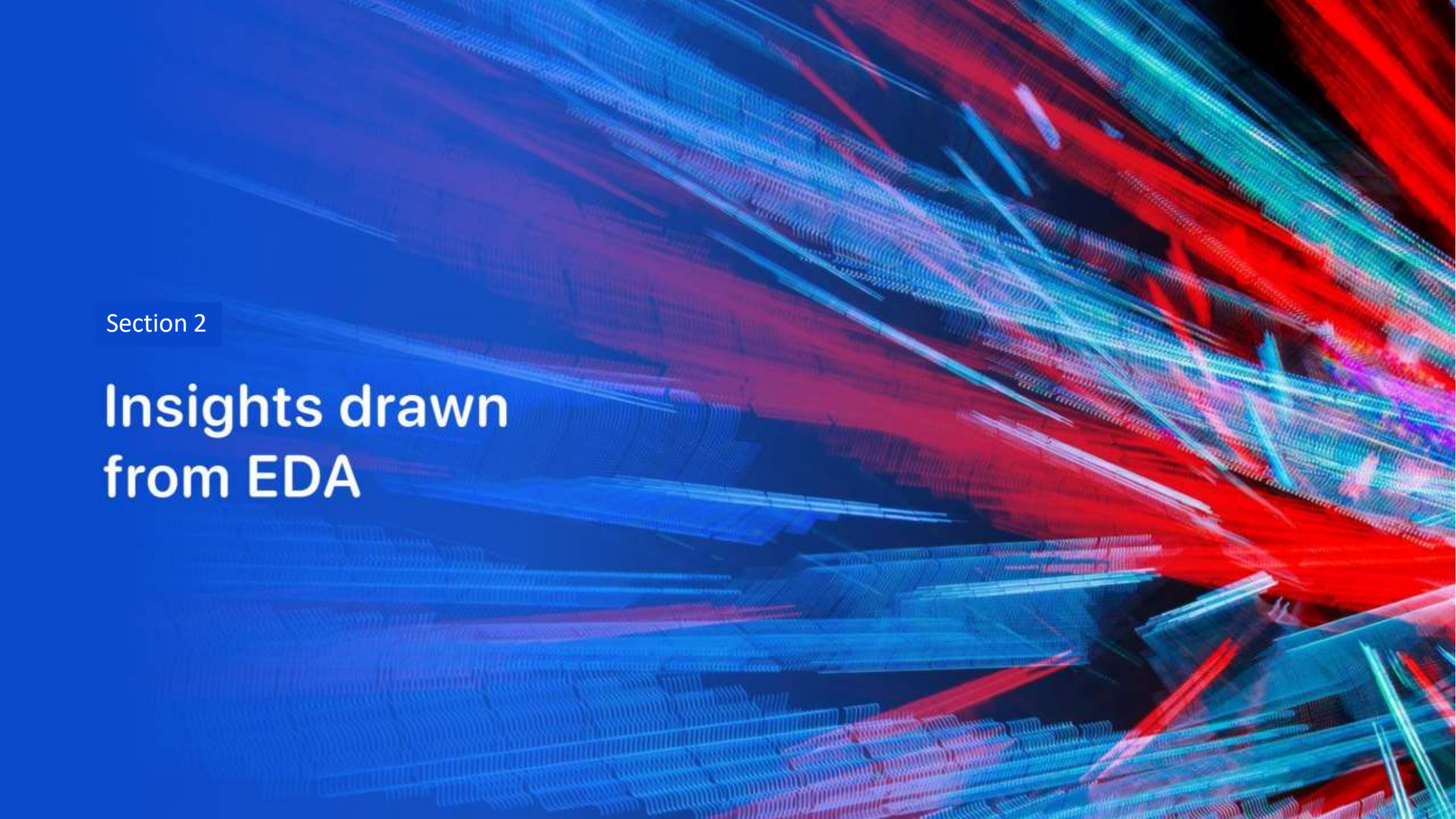
- KSC LC-39A and VAFB SLC 4E has a success rate of 77%
- VAFB SLC 4E has no payload above 10000 kg
- In the LEO orbit the Success appears related to the number of flights
- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS
- The success rate since 2013 kept increasing till 2020

## Interactive analytics



## Predictive analysis

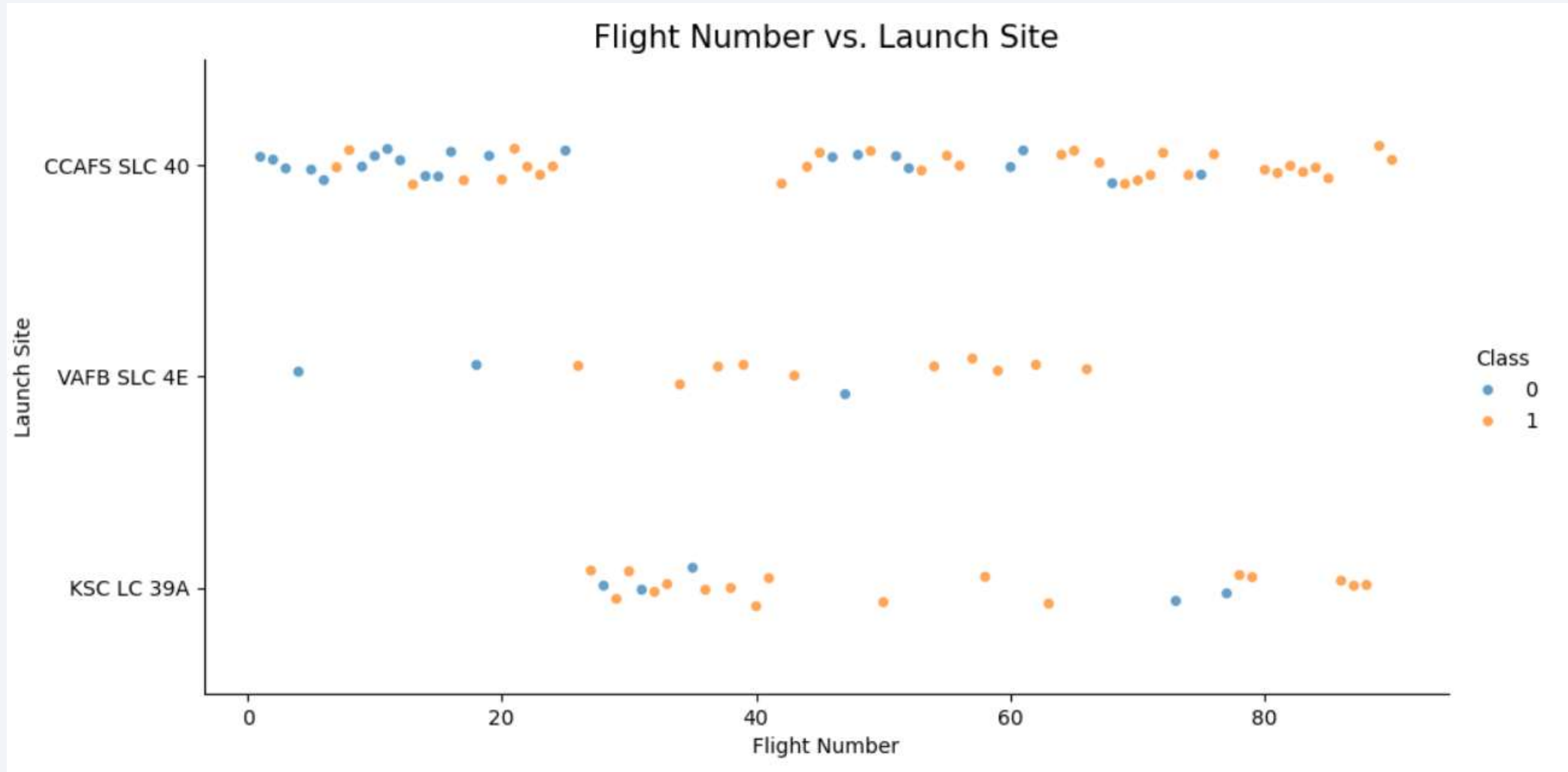
According to the decision tree classifier model, the predictive model tells us that there will be 4 true positive, 7 true negative, 5 false positive and 2 false negative. The accuracy of the model is around 89% with the best parameters.

The background of the slide is an abstract composition of numerous thin, overlapping lines and streaks in shades of blue, red, and cyan. These lines are oriented diagonally, creating a sense of motion and depth. The overall effect is a vibrant, digital-looking texture.

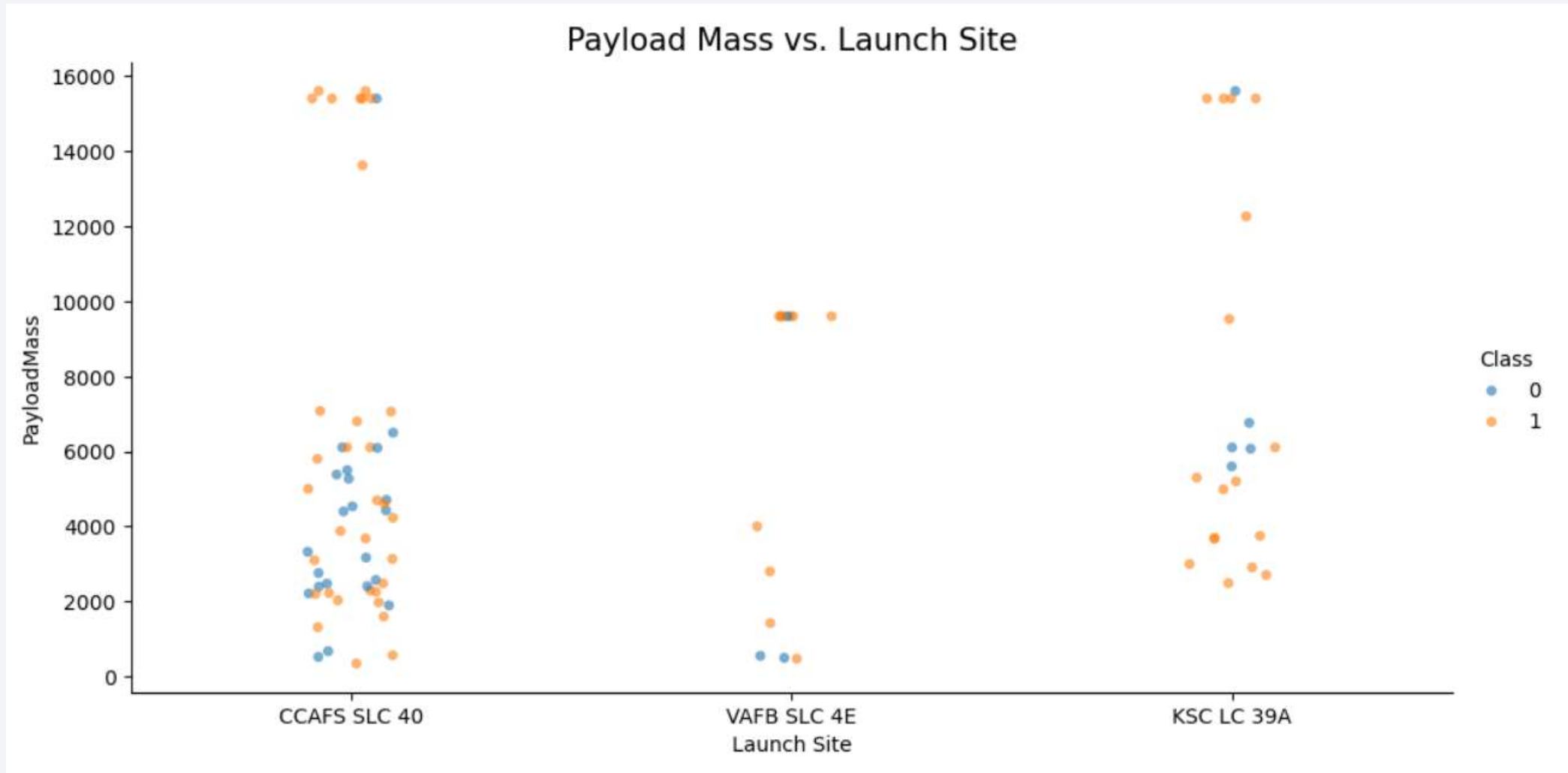
Section 2

# Insights drawn from EDA

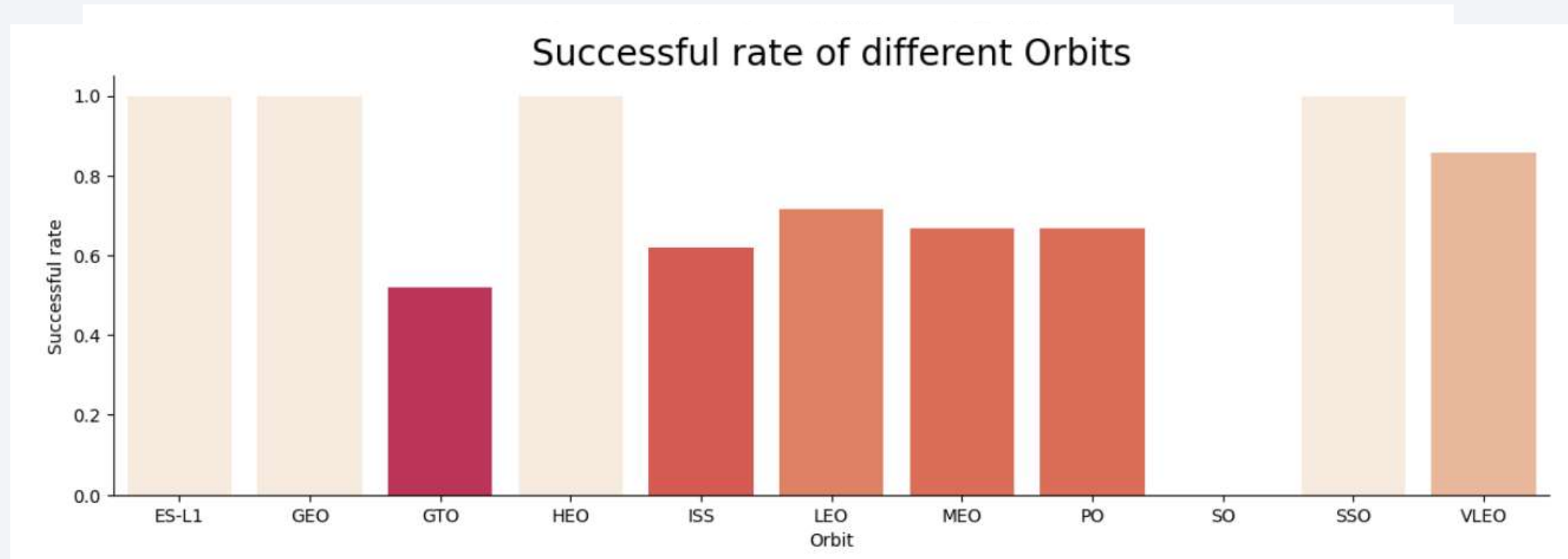
# Flight Number vs. Launch Site



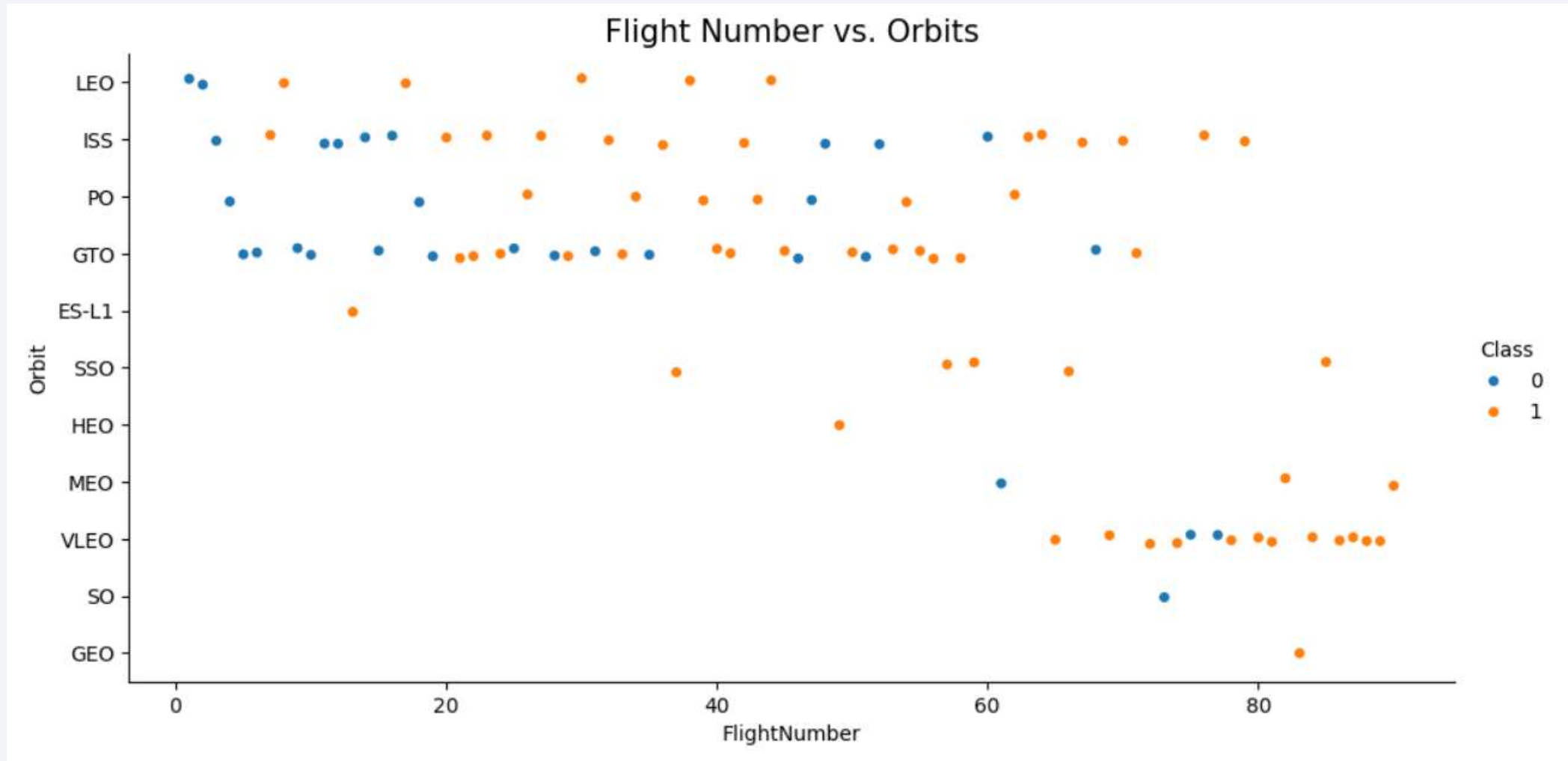
# Payload vs. Launch Site



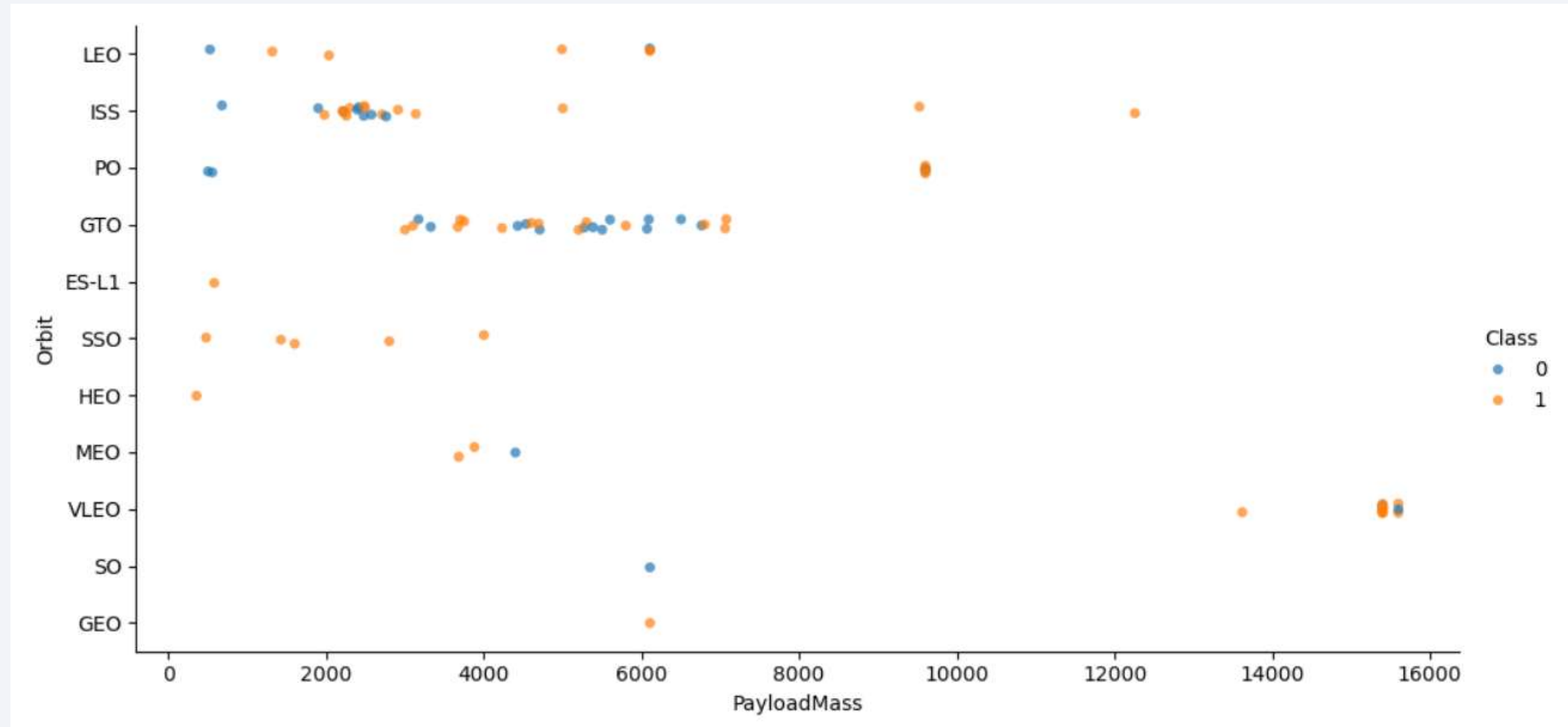
# Success Rate vs. Orbit Type



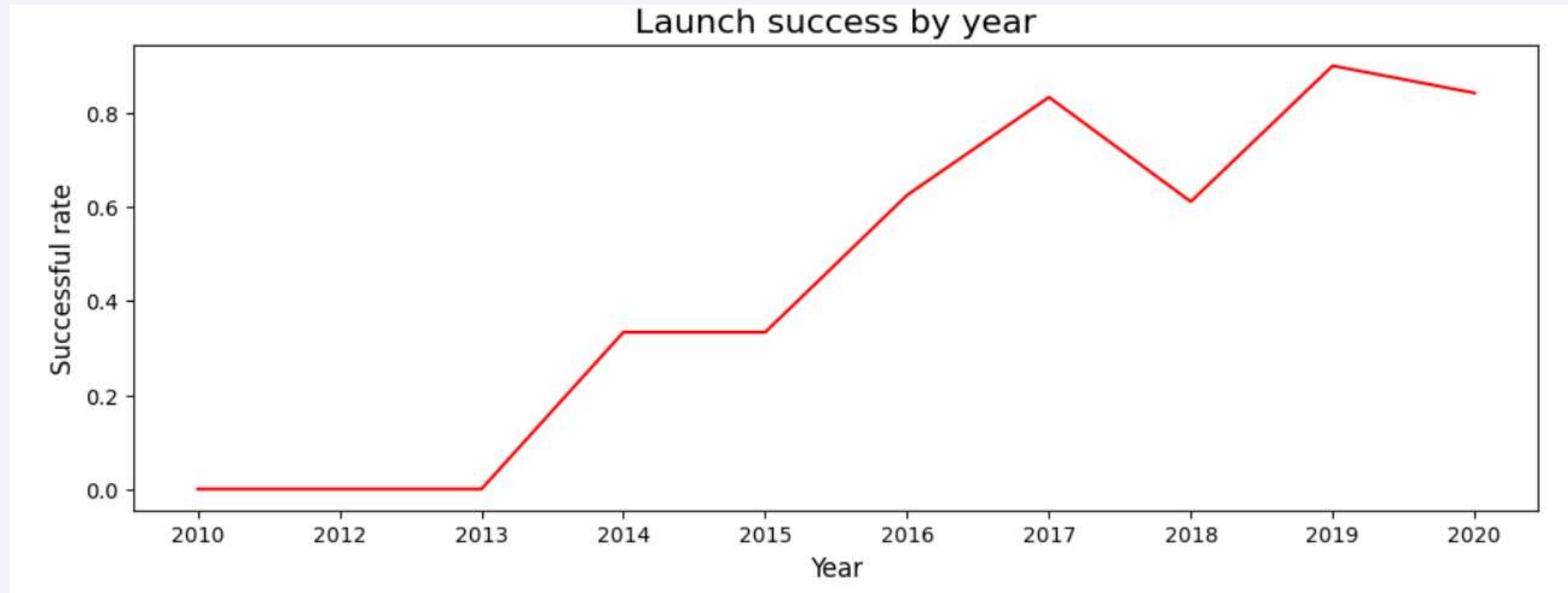
# Flight Number vs. Orbit Type



# Payload vs. Orbit Type



# Launch Success Yearly Trend



# All Launch Site Names



Display the names of the unique launch sites in the space mission

In [10]: `%sql SELECT DISTINCT Launch_Site FROM SPACEXTBL;`

`* sqlite:///my_data1.db`  
Done.

Out[10]: **Launch\_Site**

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

# Launch Site Names Begin with 'CCA'



## Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
In [11]: %sql SELECT Launch_Site FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[11]: Launch_Site
```

```
CCAFS LC-40
```

```
CCAFS LC-40
```

```
CCAFS LC-40
```

```
CCAFS LC-40
```

```
CCAFS LC-40
```

# Total Payload Mass



## Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
In [12]: %sql SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload FROM SPACEXTBL WHERE Customer LIKE 'NASA (CRS)';
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[12]: total_payload  
         45596
```

# Average Payload Mass by F9 v1.1



## Task 4

Display average payload mass carried by booster version F9 v1.1

```
In [13]: %sql SELECT avg(PAYLOAD_MASS__KG_) AS Avg_Payload FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[13]: Avg_Payload
```

```
2928.4
```

# First Successful Ground Landing Date



## Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

*Hint: Use min function*

```
In [14]: %sql SELECT min(date) AS Early_Date from SPACEXTBL where Landing_Outcome LIKE 'Success (ground pad)'
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[14]: Early_Date
```

```
2015-12-22
```

# Successful Drone Ship Landing with Payload between 4000 and 6000



## Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
In [15]: %sql SELECT DISTINCT Customer, Landing_Outcome,PAYLOAD_MASS_KG_ FROM SPACEXTBL WHERE Landing_Outcome ='Success (drone ship)' AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000
```

```
* sqlite:///my_data1.db
```

Done.

```
Out[15]:
```

|  | Customer               | Landing_Outcome      | PAYLOAD_MASS_KG_ |
|--|------------------------|----------------------|------------------|
|  | SKY Perfect JSAT Group | Success (drone ship) | 4696             |
|  | SKY Perfect JSAT Group | Success (drone ship) | 4600             |
|  | SES                    | Success (drone ship) | 5300             |
|  | SES EchoStar           | Success (drone ship) | 5200             |

# Total Number of Successful and Failure Mission Outcomes



## Task 7

List the total number of successful and failure mission outcomes

In [16]: `%sql SELECT Mission_Outcome, Count(*) AS Numbers FROM SPACEXTBL GROUP BY Mission_Outcome;`

\* sqlite:///my\_data1.db

Done.

Out[16]:

| Mission_Outcome                  | Numbers |
|----------------------------------|---------|
| Failure (in flight)              | 1       |
| Success                          | 98      |
| Success                          | 1       |
| Success (payload status unclear) | 1       |

# Boosters Carried Maximum Payload



## Task 8

List all the booster\_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
In [17]: %sql SELECT Booster_Version, Max_Payload FROM (SELECT Booster_Version, MAX(PAYLOAD_MASS__KG_) AS Max_Payload FROM SPACEXTBL GROUP BY Booster_Version)
* sqlite:///my_data1.db
Done.
```

```
Out[17]:
```

| Booster_Version | Max_Payload |
|-----------------|-------------|
| F9 B4 B1039.2   | 2647        |
| F9 B4 B1040.2   | 5384        |
| F9 B4 B1041.2   | 9600        |
| F9 B4 B1043.2   | 6460        |
| F9 B4 B1039.1   | 3310        |
| F9 B4 B1040.1   | 4990        |
| F9 B4 B1041.1   | 9600        |
| F9 B4 B1042.1   | 3500        |
| F9 B4 B1043.1   | 5000        |
| F9 B4 B1044     | 6092        |
| F9 B4 B1045.1   | 362         |

# 2015 Launch Records



## Task 9

List the records which will display the month names, failure landing\_outcomes in drone ship ,booster versions, launch\_site for the months in year 2015.

**Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.**

In [19]: `%sql SELECT SUBSTR(Date,4,2) AS Month, Booster_Version, Launch_site FROM SPACEXTBL WHERE Landing_Outcome LIKE 'Failure%drone%' AND SUBSTR(Date,`

`* sqlite:///my_data1.db`

Done.

| Month | Booster_Version | Launch_Site |
|-------|-----------------|-------------|
| 01    | F9 v1.1 B1012   | CCAFS LC-40 |
| 04    | F9 v1.1 B1015   | CCAFS LC-40 |

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20



## Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

In [20]:

```
%sql SELECT Landing_Outcome, COUNT(*) AS Numbers FROM SPACEXTBL WHERE Landing_Outcome LIKE 'Success%' AND Date BETWEEN '04-06-2010' AND '20-03-2017'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

| Landing_Outcome      | Numbers |
|----------------------|---------|
| Success              | 20      |
| Success (drone ship) | 8       |
| Success (ground pad) | 6       |

A satellite view of Earth from space, showing the curvature of the planet and the glowing city lights of the Eastern United States and parts of Canada at night. The background is a deep blue gradient.

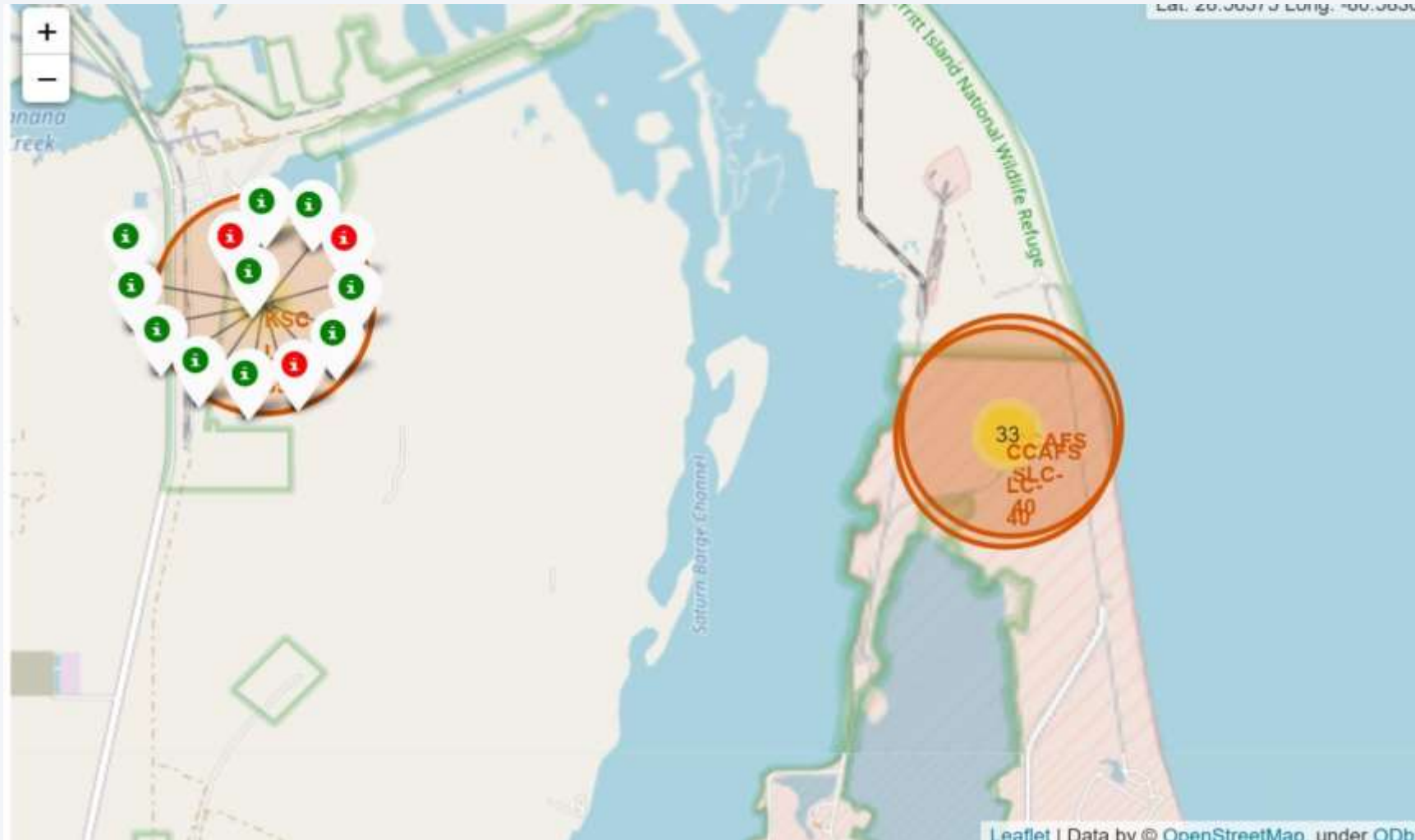
Section 3

# Launch Sites Proximities Analysis

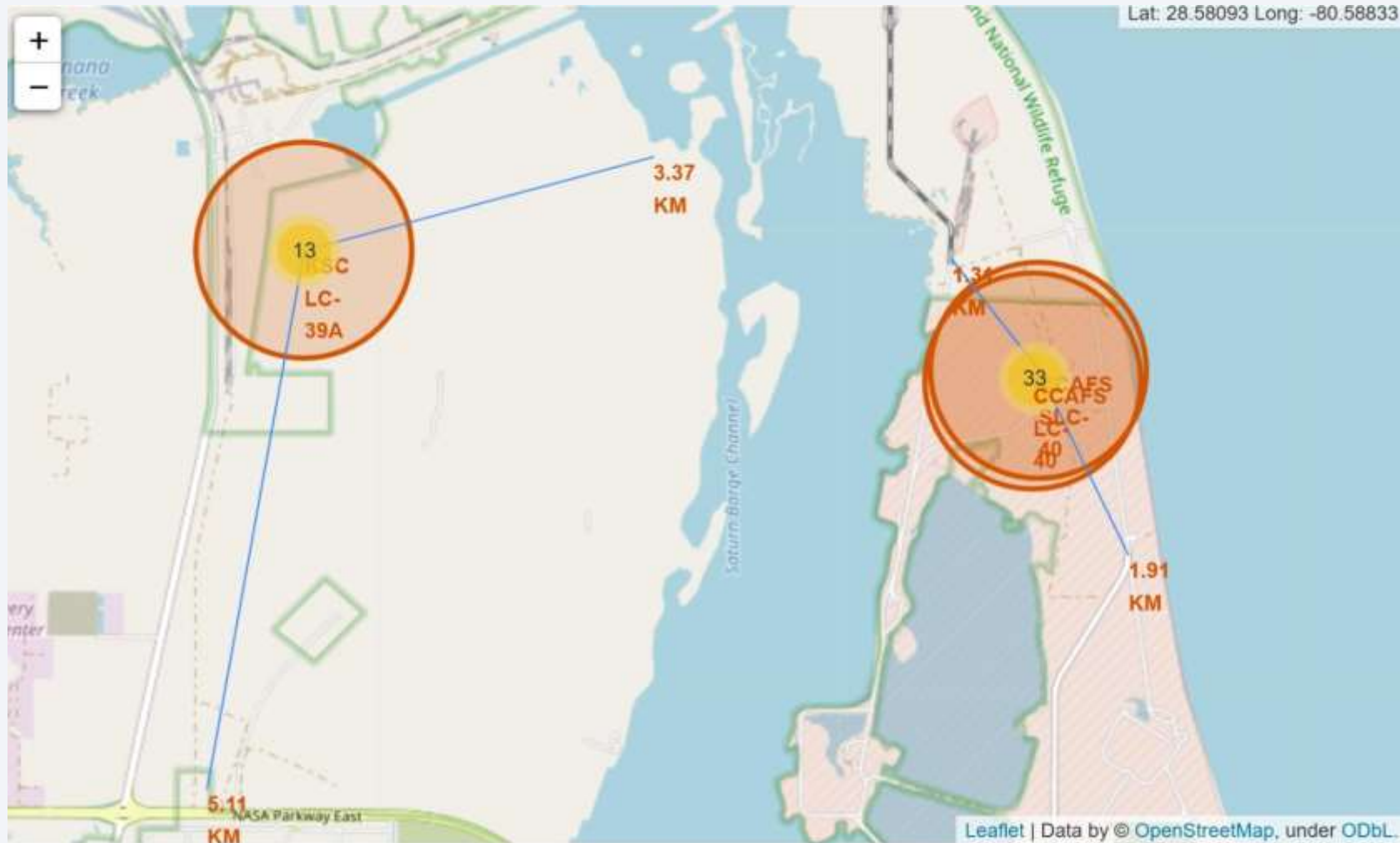


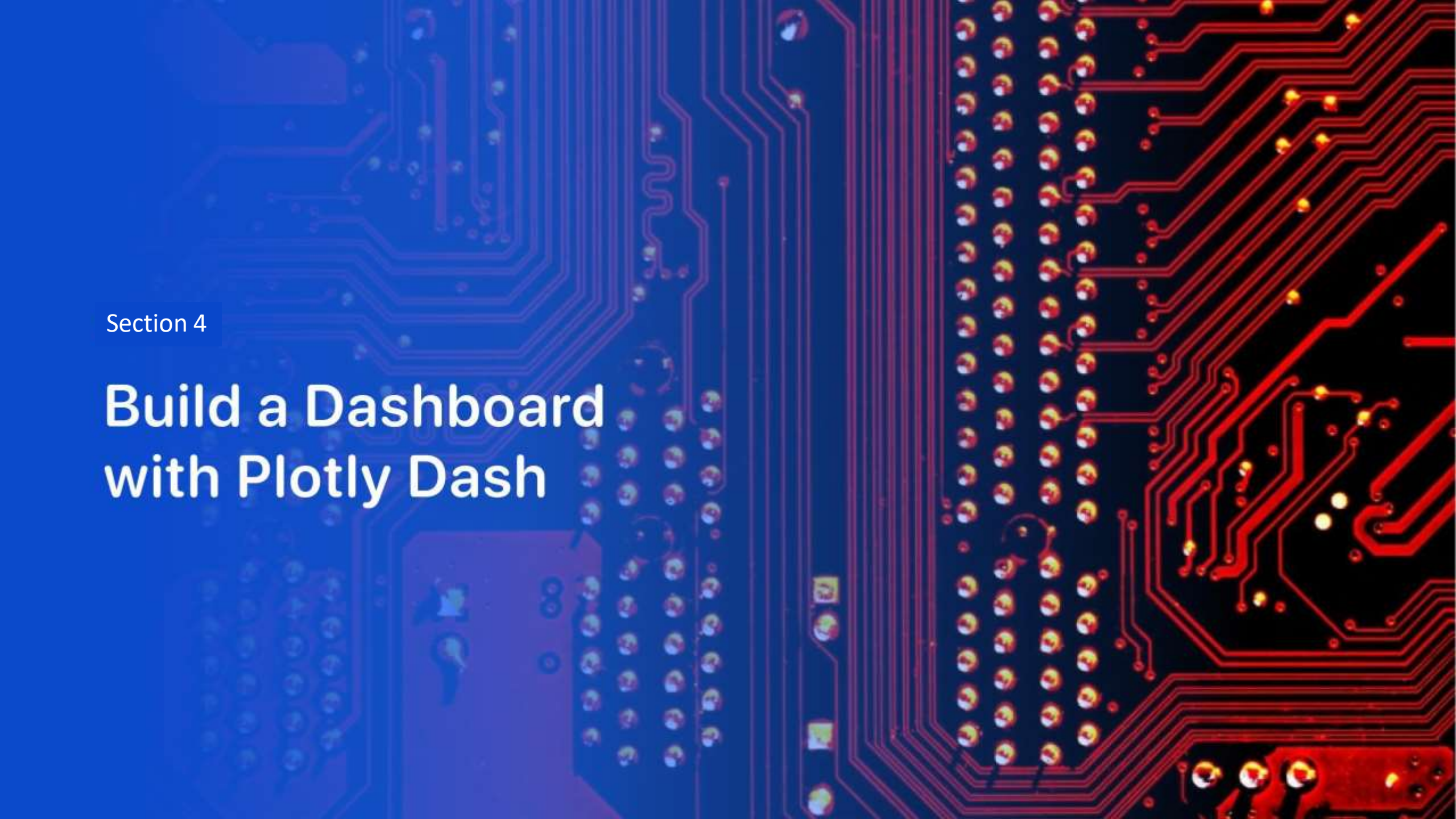
A map of the United States and Mexico. The map shows state and national boundaries. Major cities are labeled. In California, three locations are marked with orange dots and labeled: VAFB, SLC, and 4E, all near Los Angeles. In Florida, three locations are marked with orange dots and labeled: ESASF, BCC, and 80A, all near Miami. The map also shows Mexico with state boundaries and major cities like Mexico City, Monterrey, and Tijuana. The Gulf of Mexico and the Atlantic Ocean are visible. A zoom control is in the top left corner.

# Launch outcome of different site



# The proximity of the launch sites

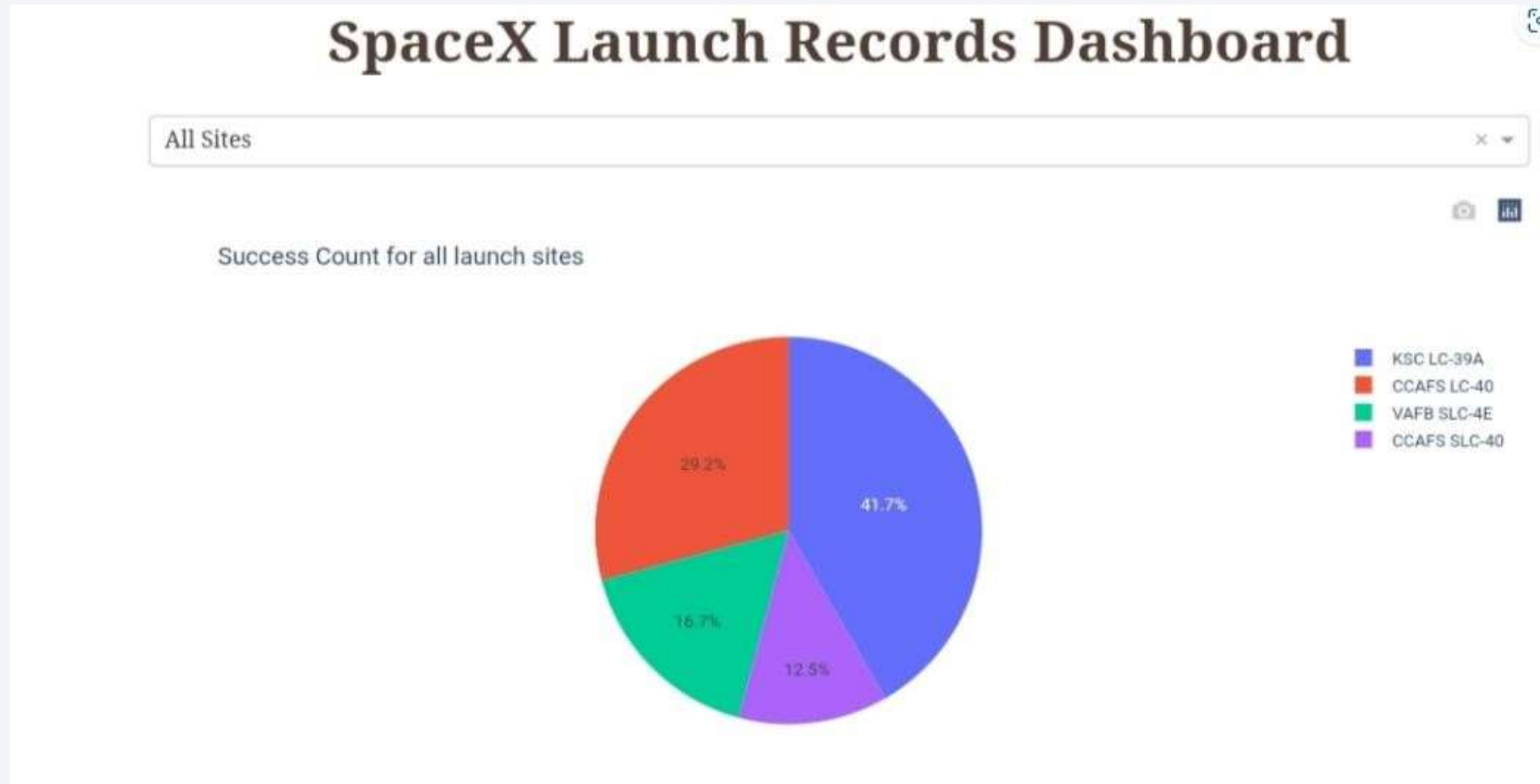




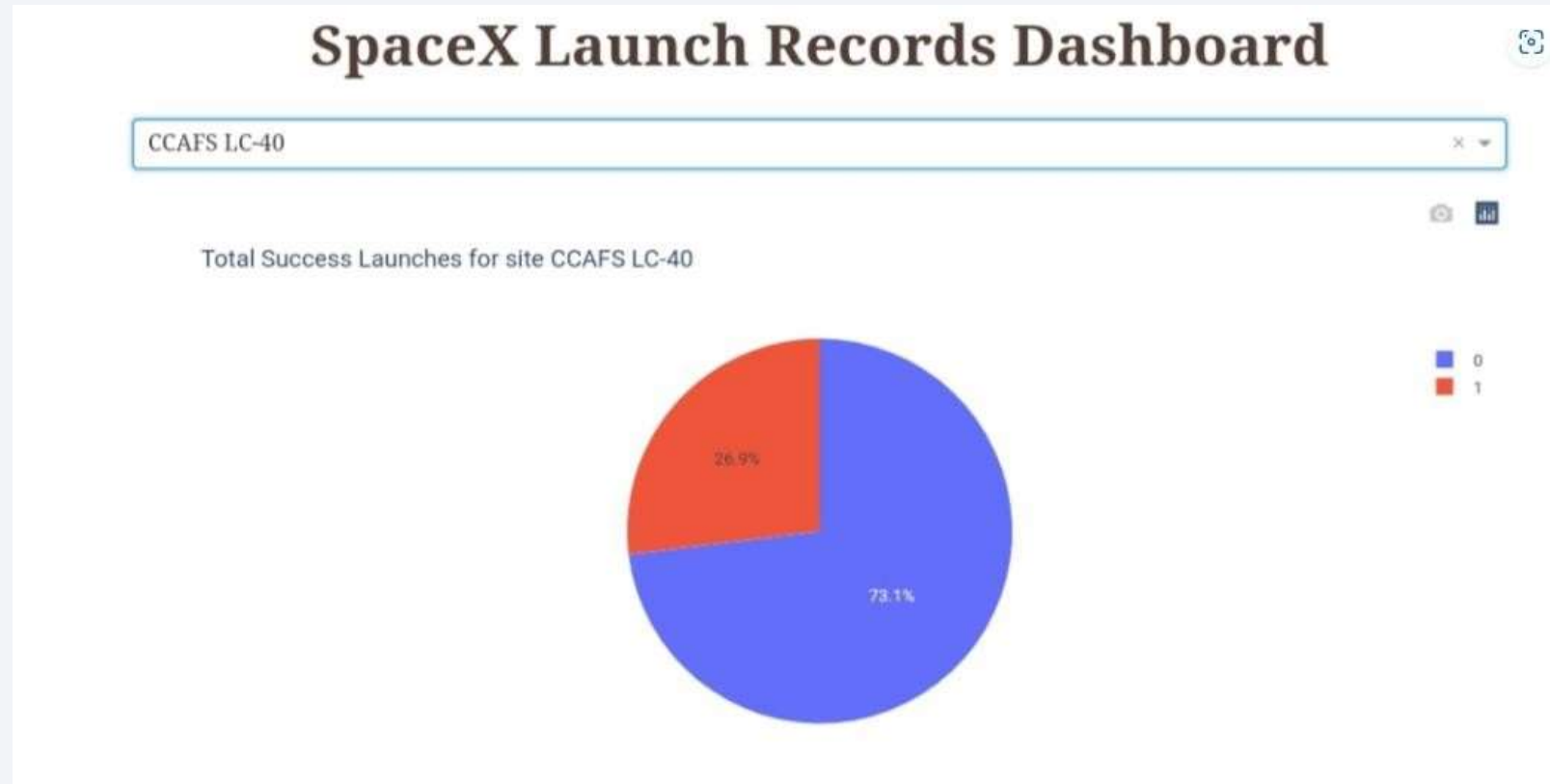
Section 4

# Build a Dashboard with Plotly Dash

# All site launch



# Highest success launch ratio





Section 5

# Predictive Analysis (Classification)

# Classification Accuracy



## TASK 12

Find the method performs best:

```
In [36]: print('Accuracy for Logistics Regression method:', logreg_cv.score(X_test, Y_test))
        print('Accuracy for Support Vector Machine method:', svm_cv.score(X_test, Y_test))
        print('Accuracy for Decision tree method:', tree_cv.score(X_test, Y_test))
        print('Accuracy for K nearsdt neighbors method:', knn_cv.score(X_test, Y_test))
```

```
Accuracy for Logistics Regression method: 0.8333333333333334
Accuracy for Support Vector Machine method: 0.8333333333333334
Accuracy for Decision tree method: 0.8888888888888888
Accuracy for K nearsdt neighbors method: 0.8333333333333334
```

- There is a correlation between launch site and success rate Payload mass is also associated with the success rate.: the more massive the payload, the less likely the first stage will return
- For orbit type, SO has the least success rate while ES-L1, GEO, HEO and SSO have the highest success rate According to the yearly trend
- There has been an increase in the success rate since 2013 kept increasing till 2020
- With best parameter provided, decision tree classifier used in prediction yielded the highest accuracy of 89%. .

# Appendix

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- [https://github.com/Jananijana2712/SpaceX-Falcon\\_9\\_First\\_Stage\\_Landing\\_Prediction](https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction)
- <https://www.coursera.org/learn/applied-data-science-capstone/home/welcome>

Thank you!

